

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-2201

Palmdale 138 kV
November 28, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2201. The Interconnection Customer proposes a 205.1 MW battery energy storage facility interconnecting at the Palmdale 138 kV substation in the New Jersey area. The request was received on October 10, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 430 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$681,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The W YATES TAP – JEMEZ JMEC 138 KV L is overloaded to 117.1% of its 208 MVA emergency rating for the loss of the VEC JAY – NORTHWEST JACKSON 345 KV. The project contributes 7.1% DFAX (14.5 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
W YATES TAP – JEMEZ JMEC 138 KV line	Rebuild line with higher-capacity conductor	\$3,443,000	20% (\$681,000)
	Total allocated to project		\$681,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.